Mohamed Zeina

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Experience

Embedded Systems & Software Engineer (Contract) — August 2025 – May. 2026
Energco, ZAIM Teknopark — *Istanbul, TR*

- Built a multi-drone system coordinating 5 UAVs with real-time 30 FPS video streaming, live telemetry via WebSockets, and fault-tolerant reconnection and failsafe handling
- Developed a Ground Control Station using React and Python FastAPI with real-time map interface, manual flight controls, and remote GPS navigation
- Implemented YOLOv8 object detection with PID-controlled flight for real-time subject tracking; calculated target GPS from gimbal angles, GPS, and rangefinder data to autonomously dispatch mini drones
- Configured bridged wireless network between GCS and drones; set up embedded Linux on Raspberry Pi for video streaming and MAVLink communication

Teaching Assistant for Introduction to Computing — Sep. 2021 – Jan. 2024
Sabancı University — *Istanbul, TR*

- Tutored over 50 students weekly in C++ programming; designed and evaluated 30+ assignments and exams aligned with course learning objectives

Full Stack Software Engineer Intern — July 2019 – Sep. 2019
Interact Technologies — *Istanbul, TR*

- Implemented TCP/IP protocol achieving 99.9% data transmission success and reduced GUI load time by 40% through code optimization
- Designed a NoSQL document database schema supporting 500+ patient records, enabling downstream reporting

Education

Sabancı University — Istanbul, TR
Master of Science in Computer Science and Engineering (M.Sc.CSE) — *Sep. 2021 – Jan. 2024*

- GPA: 3.91/4.0 | Full Scholarship | Thesis: Built a framework combining GANs, SMOTE, and differential privacy to preserve household privacy in smart meter data

Sabancı University — Istanbul, TR
Bachelor of Computer Science and Engineering (B.Sc.CSE) — *Sep. 2016 – June 2020*

- GPA: 3.73/4.0 | Half Tuition Scholarship | Degree of High Honor (2016–2017, 2018–2020)

Projects

Node Social | GitHub | *Node.js, Express.js, MongoDB, GraphQL, JWT, Cloudinary, React*

- Built a GraphQL API with JWT authentication, server-side input validation, paginated post feeds, and image upload via Multer and Cloudinary
- Implemented ownership-enforced CRUD mutations, time-limited tokens with automatic client-side expiry, and real-time communication with Socket.io

Discuss App | GitHub | *Next.js, PostgreSQL, NextAuth, NextUI, Prisma, Zod*

- Built and deployed a full-stack discussion platform with nested comment threads, GitHub OAuth, and server-side form validation with Zod
- Used Next.js App Router with React Server Components, Suspense streaming, and request memoization to deduplicate DB queries within a render

Online Shop | GitHub | *Node.js, Express.js, MongoDB, EJS, Stripe, Cloudinary, PDFKit*

- Built a full-stack MVC e-commerce app with session-based auth, bcrypt password hashing, CSRF protection, and role-scoped admin controls for product management with Cloudinary image storage
- Integrated Stripe Checkout for payments and PDFKit for server-side invoice generation; implemented a live cart using the Fetch API with real-time quantity controls, badge counter, and toast feedback

JSNote | GitHub | *React, TypeScript, ESBuild WASM, Redux, Immer, Express, Lerna, CLI*

- Published `jsnote-zeina` to npm as a CLI tool; runs a local Express server serving a React-based JavaScript notebook app
- Implemented in-browser bundling via esbuild WASM with dynamic npm resolution from unpkg.com, sandboxed iframe execution, and cumulative cell scope sharing across the notebook

Skills

Languages: C++, C#, Python, Java, JavaScript, TypeScript, SQL
Frameworks & Libraries: React, Redux, Next.js, Node.js, Express, Mongoose, GraphQL, Prisma, Socket.io
Web & Databases: Tailwind CSS, RESTful APIs, WebSockets, MongoDB, PostgreSQL, SQLite, Redis
DevOps: Docker, Nginx, Kubernetes, AWS (EC2, RDS, S3), Google Cloud (GKE), CI/CD (GitHub Actions, Travis CI)
Embedded & CV: Raspberry Pi, MAVLink, UDP Streaming, OpenCV, YOLOv8
Spoken: Arabic (Native), English (Fluent)